

GenAI : Principles and Applications

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"I was just like you few days before , I just sat , read , gave some time , understood things from their first principles and here I'm with you , among you and will be with you – Im not special , Im still just like you with some time invested in something."

(Nagesh)

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This is the outline of 12 week course on generative AI that we are going to cover in the Aug-Dec 2026 semester.

Course Outline

We start with classical machine learning and introduce the idea of distribution estimation. We formulate the fundamental problem of machine learning.

After that we will start to discuss modern methods using neural networks. First we build a neural network without libraries where we discuss methods such as back propagation , regularization techniques , normalization.

After we will jump to build latent models such as AE and VAE We then introduce GANs. We turn finally to Sequence modelling where we cover generalized RNNs , LSTMs, GRUs. Attention based Transformers.

Finally we teach diffusion and Flow models include DDPM,DDIM,RF and NF.

If time permits we will cover introductory ideas of RL.

Couser syllabus: Week By week

Here we provide the syllabus of what we are going to cover in 12 weeks.

- Week 01 : Introduction to Machine Learning and distribution estimation.
- Week 02 : Introduction to neural networks and building one from scratch without libraries.
- Week 03 : Introduction to latent variable models AutoEncoder and Variational Auto encoder
- Week 04 : How we use AE and VAE ?(Practical session)
- Week 05 : GANs models
- Week 06 : How we use GANs ?(Practical session)
- Week 07 : Sequence Models : From RNN to Transformer
- Week 08 : Practical sessions on Sequence Models
- Week 09 : Diffusion and Flow Models(Theory)
- Week 10 : Practical sessions on Diffusion and Flow
- Week 11 : Introduction to RL
- Week 12 : Revision and conclusions